‘banker’ had magically appeared, making it possible for both Bristol and Coventry to benefit from the same outcome, a draw.

Spectator sports like football are normally zero sum games for a good reason. It is more exciting for crowds to watch players striving mightily against one another than to watch them conniving amicably. But real life, both human life and plant and animal life, is not set up for the benefit of spectators. Many situations in real life are, as a matter of fact, equivalent to nonzero sum games. Nature often plays the role of ‘banker’, and individuals can therefore benefit from one another’s success. They do not have to do down rivals in order to benefit themselves. Without departing from the fundamental laws of the selfish gene, we can see how cooperation and mutual assistance can flourish even in a basically selfish world. We can see how, in Axelrod’s meaning of the term, nice guys may finish first.

But none of this works unless the game is *iterated*. The players must know (or ‘know’) that the present game is not the last one between them. In Axelrod’s haunting phrase, the ‘shadow of the future’ must be long. But how long must it be? It can’t be infinitely long. From a theoretical point of view it doesn’t matter how long the game is; the important thing is that neither player should *know* when the game is going to end. Suppose you and I were playing against each other, and suppose we both knew that the number of rounds in the game was to be exactly 100. Now we both understand that the 100th round, being the last, will be equivalent to a simple one-off game of Prisoner’s Dilemma. Therefore the only rational strategy for either of us to play on the 100th round will be DEFECT, and we can each assume that the other player will work that out and be fully resolved to defect on the last round. The last round can therefore be written off as predictable. But now the 99th round will be the equivalent of a one-off game, and the only rational choice for each player on this last but one game is also DEFECT. The 98th round succumbs to the same reasoning, and so on back. Two strictly rational players, each of whom assumes that the other is strictly rational, can do nothing but defect if they both know how many rounds the game is destined to run. For this reason, when games theorists talk about the Iterated or Repeated Prisoner’s Dilemma game, they always assume that the end of the game is unpredictable, or known only to the banker.

Even if the exact number of rounds in the game is not known for certain, in real life it is often possible to make a statistical guess as to